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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Specialized Project – Artificial Intelligence

HK261-DAGD1-041

Technical Report

An End-to-End MLOps Pipeline for Cross-Sectional Equity Ranking under Data Constraints: Engineering Validation and Ablation Study.

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HO CHI MINH CITY, 3 NOVEMBER 2026

Declaration

We, the undersigned – Doan Phuong Hung Cuong, Nguyen Thien Hao, and Pham Quang Hieu – hereby submit this Specialized Project Report, titled *"An End-to-End MLOps Pipeline for Cross-Sectional Equity Ranking under Data Constraints: Engineering Validation and Ablation Study"*, in partial fulfillment of the academic requirements. We solemnly declare that this manuscript represents the authentic outcome of our independent research, system architecture design, and empirical validation, conducted under the supervision of Dr. Pham An Vinh and Dr. Nguyen An Khuong.

We explicitly affirm the following statements regarding academic integrity, ethical standards, and original authorship:

- **Originality and Intellectual Attribution:** All methodologies, algorithms, data sources, and literature referenced herein that are not our original creation have been explicitly acknowledged, formally cited, and referenced in strict compliance with established academic standards.
- **Code Ownership and Reproducibility:** All software artifacts, system architectures, and codebase implementations were engineered by the authors. Any third-party libraries, open-source frameworks, or cloud infrastructures utilized within the pipeline have been properly cited and applied in accordance with their respective licenses.
- **Disclosure of AI and LLMs Assistance:** Generative Artificial Intelligence (AI) systems and Large Language Models (LLMs) were utilized strictly as supportive tools for syntactic refinement, prose editing, and structural organization. No core technical architectures, empirical analyses, feature engineering logic, or conclusions were synthesized by automated tools. The authors retain full ownership of and accountability for all intellectual content.
- **Data Integrity and Empirical Veracity:** All computational experiments, performance metrics, and ablation outcomes presented in this work reflect genuine implementation results without fabrication, falsification, or deliberate omission of data.
- **Exclusivity of Submission:** This research work is original and has not been submitted, either in whole or in part, for the award of any other academic degree, diploma, or professional qualification at this or any other institution.

Ho Chi Minh City, 3rd November 2026

Doan Phuong Hung Cuong	Nguyen Thien Hao	Pham Quang Hieu

Acknowledgment

First and foremost, we extend our deepest gratitude to Dr. Pham An Vinh for his outstanding technical mentorship and visionary guidance throughout this project. His expertise in quantitative finance and practical insights into alpha strategies were central to our research. His real-world experience helped us address the complexities of modeling financial markets, recognizing statistical trends, and analyzing microstructure dynamics. Moreover, his rigorous supervision greatly enhanced the precision of our quantitative analysis, including empirical evaluation, feature engineering, and backtesting of our equity ranking pipelines.

We also wish to express our sincere appreciation to our supervisor, Dr. Nguyen An Khuong. His invaluable support and steadfast academic guidance have been instrumental in refining our technical acumen and instilling a rigorous, systematic approach to both our research and engineering efforts. We are especially thankful for his generous provision of foundational academic materials and specialized literature, which proved critical to the successful formulation, structural organization, and execution of our research objectives.

We also extend our earnest appreciation to Mr. Nguyen Duong Nhat (M.Sc. Candidate, FAU) and Miss Truong Nguyen Duyen Anh (B.A. Candidate, Universität Augsburg) for their invaluable assistance in sourcing and compiling the datasets utilized in this thesis. Additionally, we are deeply grateful to Mr. Nguyen Khac Khanh Huy (B.A. Candidate, HCMUT) for his vital contributions in engineering and constructing the robust data warehouse infrastructure. Their collective efforts in data acquisition, preparation, and architectural design provided the essential technical foundation for the empirical analysis and validation of our quantitative models.

Our sincere thanks are directed to the Faculty of Computer Science and Engineering at Ho Chi Minh City University of Technology for fostering a stimulating academic environment and supplying the foundational resources requisite for conducting this specialized study.

Furthermore, we gratefully acknowledge Modal for their generous provision of cloud computing infrastructure. Access to their high-performance, complimentary Graphics Processing Unit resources was pivotal in overcoming our local hardware limitations, enabling the efficient execution of complex computational experiments, large-scale ablation studies, and the training of our machine learning pipelines.

We also wish to recognize our peers and colleagues whose intellectual camaraderie, constructive feedback, and steadfast encouragement provided much-needed motivation during the more demanding phases of this project.

Finally, we are eternally grateful to our families for their unwavering emotional support, patience, and understanding throughout the extensive development, implementation, and writing of this report.

Abstract

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List of Abbreviations

AI Artificial Intelligence

API Application Programming Interface

DSMA Distance from Simple Moving Average

GPU Graphics Processing Unit

IC Information Coefficient

ICIR Information Coefficient Information Ratio

LLMs Large Language Models

LSTM Long Short-Term Memory

MLOps Machine Learning Operations

NYSE New York Stocks Exchange

OHLCV Open-High-Low-Close-Volume

PVT Price Volume Trend

RSI Relative Strength Index

Part I

Introduction

1 Context and Motivation

Machine learning has become an increasingly important methodological tool in quantitative finance because of its ability to model nonlinear relationships and interactions among large sets of financial predictors. In a large-scale empirical study of equity returns, Gu, Kelly, and Xiu [GKX20] show that machine learning methods can improve return forecasting relative to conventional linear approaches, with part of the predictive improvement arising from nonlinear interactions among characteristics. Their results also identify momentum, liquidity, and volatility-related variables among the dominant predictors of expected returns. This evidence motivates the use of flexible nonlinear models when the objective is to extract information from a rich set of financial signals.

Sequence-based deep learning models provide one approach to capturing temporal structure in financial data. Fischer and Krauss [FK18] demonstrate the application of Long Short-Term Memory (LSTM) networks to financial market prediction and report that LSTM models can exploit patterns in sequential stock data that are difficult to capture using simpler predictive models. More generally, the Transformer architecture [VSP+17] provides a mechanism for modeling relationships through attention, allowing a model to selectively incorporate information from multiple elements of an input sequence. These characteristics make recurrent and attention-based architectures potentially suitable for financial problems in which relevant information may be distributed across time and across simultaneously observed securities.

The present study focuses specifically on the latter dimension. The objective is not to forecast the return of an individual stock in isolation, but to generate a *cross-sectional ranking* of securities at each decision point. This distinction is fundamental because the economic decision in a cross-sectional strategy is based on the relative ordering of securities rather than the numerical accuracy of an isolated return forecast. Poh et al. [PLZR21] formalize this perspective and show that ranking-oriented learning methods can improve cross-sectional systematic strategies by directly learning pairwise and listwise relationships among instruments. The ranking problem therefore requires a modeling framework that preserves information about the contemporaneous cross-section instead of treating each security as an independent forecasting task.

The cross-sectional nature of equity returns is also well established in the asset-pricing literature. Fama and French [FF92] show that firm characteristics such as size and book-to-market equity are related to systematic differences in average stock returns, while Fama and French [FF93] demonstrate that common risk factors account for substantial shared variation in stock returns.

These findings motivate a distinction between systematic variation shared across securities and residual variation that may contain additional information about relative performance. In the present study, this distinction is reflected in the construction of a forward-return target that is residualized against common risk factors before cross-sectional evaluation.

The research problem is therefore not simply to determine whether a flexible model can predict future stock returns. Rather, it is to determine whether temporal information and cross-sectional relationships can be combined to improve the ranking of securities within a common market universe. The proposed architecture addresses this problem by combining an LSTM-based temporal encoder with a cross-sectional Transformer, allowing information about the historical evolution of an individual security to be processed jointly with information from the contemporaneous cross-section.

2 Research Problem

The central research problem addressed in this thesis is the design and empirical validation of an end-to-end machine learning pipeline for cross-sectional equity ranking under the statistical, data, and implementation constraints inherent in financial markets.

A conventional pointwise forecasting formulation assigns a prediction problem to each security separately. However, the investment decision considered in this study is inherently comparative: securities are ranked relative to their peers and subsequently incorporated into a portfolio according to their predicted relative performance. The learning objective must therefore reflect the structure of the downstream decision problem. This motivates the use of a ranking-aware learning objective in addition to the conventional regression objective, following the cross-sectional learning-to-rank perspective of Poh et al. [PLZR21].

At the same time, the predictive problem must account for common systematic variation. Because securities in the same market universe are exposed to shared market conditions, raw forward returns contain variation that may not represent security-specific information. The empirical design therefore residualizes the forward return against common factors before evaluating cross-sectional predictive performance. This approach is consistent with the role of common risk factors established in the asset-pricing literature [FF92;FF93].

A second dimension of the research problem concerns empirical validation. Financial observations are temporally ordered and may contain overlapping information across adjacent observations. Consequently, model evaluation must preserve chronological structure and distinguish information available at the prediction date from information that becomes available subsequently. Tashman [Tas00] emphasizes the importance of carefully designed out-of-sample evaluation and rolling-origin procedures in forecasting applications. In addition, extensive hy-

perparameter experimentation creates a multiple-testing problem: a model selected because it performs well over many trials may exhibit an inflated backtest performance even when the underlying strategy contains little genuine predictive information. Bailey and López de Prado[BLdP14] develop the Deflated Sharpe Ratio specifically to address selection bias and backtest overfitting arising from multiple testing.

The research therefore treats validation as part of the model itself rather than as a separate reporting step. The proposed pipeline incorporates chronological walk-forward evaluation, purging and embargoing of observations around test periods, and explicit adjustment for multiple-testing effects. These procedures are intended to reduce the possibility that apparent predictive performance results from information leakage or repeated experimentation rather than from generalizable structure.

Finally, the research considers the transition from statistical prediction to portfolio construction. A cross-sectional ranking may contain predictive information without being directly implementable as a portfolio. Portfolio decisions are affected by covariance estimation, position constraints, turnover, transaction costs, and market impact. The framework consequently evaluates the model's out-of-sample predictions within a constrained portfolio construction process. The importance of accounting for execution costs and market impact in portfolio implementation is established in the optimal-execution literature[AC00].

3 Research Gap

The first research gap concerns the alignment between prediction objectives and cross-sectional portfolio decisions. Standard regression-based forecasting evaluates the accuracy of individual return predictions, whereas cross-sectional strategies ultimately depend on the relative ranking of securities. Poh et al.[PLZR21] demonstrate the importance of learning the ranking structure directly. The present study extends this perspective by investigating a hybrid learning objective that combines pointwise prediction with pairwise ranking information within a deep temporal–cross-sectional architecture.

The second gap concerns the integration of temporal and cross-sectional information within a single predictive framework. LSTM-based models provide a mechanism for learning sequential dependencies[FK18], while attention-based models provide a mechanism for representing interactions among elements of an input set[VSP+17]. Existing empirical work establishes the usefulness of both nonlinear machine learning and cross-sectional structure in equity prediction[GKX20;PLZR21]; however, the present study focuses on an integrated architecture in which temporal representations of individual securities are subsequently processed jointly across the cross-section.

The third gap concerns validation under realistic financial data constraints. Machine learning models provide substantial flexibility, but that flexibility also increases exposure to data leakage, selection bias, and backtest overfitting. Tashman [Tas00] establishes the importance of carefully designed out-of-sample forecasting procedures, while Bailey and López de Prado [BLdP14] show that multiple testing can materially inflate apparent strategy performance. The present research incorporates these concerns directly into the experimental design through purged walk-forward validation, embargoing, and multiple-testing diagnostics.

The fourth gap concerns the connection between predictive signals and implementable portfolios. Rather than evaluating the model solely through prediction accuracy, the study carries locked out-of-sample predictions into a constrained portfolio construction stage. This allows the robustness of the predictive signal to be examined after accounting for covariance risk and explicit transaction-cost and market-impact assumptions, consistent with the principles of optimal execution [AC00].

4 Research Questions

The central research question is:

> *Can an end-to-end MLOps pipeline successfully engineer, validate, and constrain a hybrid temporal-attention (LSTM–Transformer) architecture for cross-sectional equity ranking, and accurately translate its out-of-sample predictions into a strictly risk-neutralized portfolio under rigid computational and data constraints?*

This question is investigated through two groups of research questions. The first concerns signal generation and model validation. It examines whether a hybrid loss combining Mean Squared Error (MSE) with a pairwise hinge loss improves the separation of predicted cross-sectional deciles relative to a pure pointwise objective, and whether a learned static Stock ID embedding contributes information beyond the temporal and cross-sectional market structure.

The second group concerns portfolio construction and robustness. It examines the degradation of model-derived portfolio performance after the introduction of covariance risk and transaction costs, and evaluates the extent to which multiple-testing adjustment reduces the apparent performance obtained during model and hyperparameter selection.

5 Research Contributions

The contribution of this thesis is primarily methodological and empirical. It develops and evaluates an end-to-end framework for cross-sectional equity ranking in which predictive modeling, statistical validation, and portfolio construction are treated as interconnected components.

First, The study formulates the equity prediction problem explicitly as a cross-sectional ranking task rather than as a set of independent point forecasts.

Second, It combines temporal representation learning through an LSTM with cross-sectional attention through a Transformer, thereby allowing both within-security dynamics and contemporaneous inter-security relationships to be modeled.

Third, It incorporates a ranking-aware training objective intended to align the optimization problem with the downstream portfolio decision.

Fourth, The framework embeds chronological and leakage-aware validation procedures together with explicit control for multiple-testing bias.

Finally, It evaluates the persistence of the generated signal after portfolio construction under covariance, leverage, position, turnover, and transaction-cost constraints.

The study is deliberately framed as an engineering validation and empirical evaluation rather than a claim that the resulting model constitutes universally deployable trading alpha. The empirical analysis is conducted within the explicitly defined S&P 500 research universe and its filtered analytical sample, and the methodological limitations associated with the available data are retained and discussed explicitly.

6 Thesis Organization

The remainder of the thesis follows the sequence of the empirical pipeline. The methodology first defines the market universe and constructs the analytical sample, including the associated survivorship and classification considerations. The data-processing stage then describes the construction and transformation of the input variables and prediction target. The predictive modeling stage presents the temporal and cross-sectional architecture, the ranking-aware learning objective, and the experimental design. The subsequent evaluation examines predictive performance, statistical significance, model stability, and sensitivity to alternative specifications. Finally, the locked out-of-sample predictions are passed to the portfolio-construction framework, where the economic consequences of risk controls and transaction costs are evaluated before the methodological limitations and conclusions are presented.

7 Overall Structure

To be updated

Part II

Methodology

8 Problem Formulation

The Cross-Sectional Ranking Objective

The investment decision analyzed in this study is inherently comparative: the goal is to rank securities relative to each other, rather than to predict the absolute numeric accuracy of each individual return. As such, the problem is cast as a *learning-to-rank* task. Denote by N the number of active equities in the analytical sample at time t. At each prediction date, the model receives a three-dimensional input tensor of temporal features, $\mathbf{X}_t \in \mathbb{R}^{N \times T \times F}$, where T is the lookback sequence length (e.g., 60 days), and F is the number of engineered features extracted per security.

The learning objective is to parameterize a mapping $f(\mathbf{X}_t; \Theta)$ that outputs a vector of cross-sectional scores $\hat{\mathbf{y}}_t \in \mathbb{R}^N$. These scores should monotonically align with the realized forward *residualized* returns $\mathbf{y}_t \in \mathbb{R}^N$, meaning the induced orderings serve as the ranking for the cross-section at time t.

Generalization and Bias Neutralization

A central design constraint is to prevent structural survivorship bias and regime overfitting. Since the sample is composed of continuous survivors from 2020 onward, including static identifiers (such as Stock IDs or persistent entity embeddings) would allow the model to memorize long-term "winners", thus violating generalizability. To enforce that the model learns a generalized mapping (a pricing kernel), $f(\mathbf{X}_t; \Theta)$ is constructed to explicitly exclude such static structural identifiers. Both the temporal encoder (LSTM) and the context layer (Cross-Sectional Transformer) must deduce the cross-sectional ranking solely from the time-varying technical and fundamental features present in $\mathbf{X}_t$, observed at each t.

Hybrid Optimization Function

To align mathematical training with the requirements of downstream portfolio construction, the network is trained with a *hybrid* loss. This combines classical Mean Squared Error (MSE), which helps anchor the scale of the predictions, with a pairwise hinge loss. The hinge component operates on each cross-sectional batch and explicitly penalizes situations where securities from empirically lower-performing quantiles are scored above those from higher-performing

quantiles. This approach directly incentivizes the monotonic separation required for constructing a market-neutral, cross-sectional portfolio.

9 Target Construction

Systematic Risk Residualization

Securities within a broad equity universe share exposure to the same market-wide risk factors; thus, raw forward returns are dominated by common systematic variation that is not stock-specific. To ensure that the model's learning target strictly captures *security-specific* signals, the forward-looking prediction target is residualized with respect to known risk factors prior to cross-sectional evaluation.

Fama-French 3-Factor Rolling OLS

The idiosyncratic return target for each stock is constructed via the Fama-French 3-Factor model. For every security i, a trailing 126-day (roughly six months) rolling Ordinary Least Squares (OLS) regression is performed at date t:

$$R_{i,t} - R_{f,t} = \alpha_{i,t} + \beta_{i,\text{MKT}}\,\text{MKT}_t + \beta_{i,\text{SMB}}\,\text{SMB}_t + \beta_{i,\text{HML}}\,\text{HML}_t + \epsilon_{i,t} \tag{9.1}$$

where $R_{i,t}$ is the return for stock i, $R_{f,t}$ is the risk-free rate, and MKT_t, SMB_t, and HML_t denote the market, size, and value factor premia, respectively. The residual $\epsilon_{i,t}$ isolates the stock-specific, non-systematic return component at each date.

Cross-Sectional Z-Scoring for the Forward Horizon

To produce the final set of predictive targets $\mathbf{y}_t$ for each ranking instance, the daily idiosyncratic residuals $\epsilon_{i,t}$ are aggregated for each security over a 5-day forward prediction horizon. To ensure that the resulting target is scale-invariant and purely comparative, these 5-day aggregated residuals are cross-sectionally Z-scored at each t. That is, for the vector of N stocks on a given day, the transformed target has mean 0 and standard deviation 1 across the cross-section:

$$\mathbf{y}_t = \frac{\boldsymbol{\epsilon}_t^{(5)} - \text{mean}(\boldsymbol{\epsilon}_t^{(5)})}{\text{std}(\boldsymbol{\epsilon}_t^{(5)})} \tag{9.2}$$

This enforces that the model's predictions are evaluated exclusively by their relative ordering, consistent with the long/short, market-neutral ranking objective.

10 Project Objectives

To be updated

Part III

Market Reconstruction

Survivorship bias is a well-established problem in empirical financial research: restricting historical analysis to securities that survive until the end of the sample can produce distorted estimates and misleading evidence of return predictability. Brown et al.[BGIR92] demonstrate that survivorship-truncated samples can materially affect empirical conclusions.

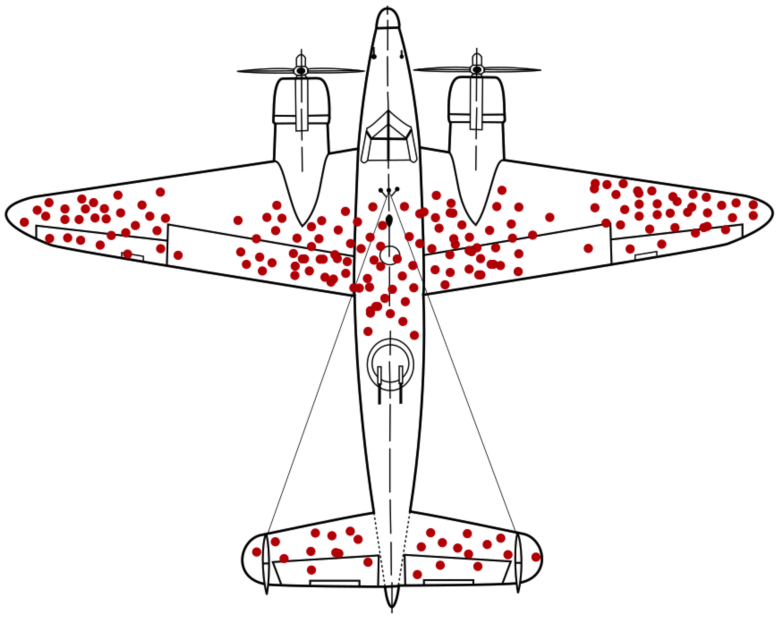


Figure 10.1: Abraham Wald's Survivorship Bias Plane

For this study, avoiding survivorship bias is complicated by the limited access provided by free-tier market-data services. No single freely accessible source is assumed to provide both a sufficiently long historical security universe and the complete security-level price data required by the study. Consequently, the dataset must be reconstructed by combining complementary sources rather than directly downloaded as a finished historical panel.

The pipeline therefore uses the New York Stocks Exchange (NYSE) trading calendar as its temporal backbone, retrieves daily historical active-ticker snapshots from Wharton Research Data Services W, aggregates these observations with Polars to derive ticker availability intervals, and uses those intervals to obtain daily OHLCV data from Yahoo Finance. This produces a persistent raw market dataset that preserves historical security availability while remaining feasible under the constraints of the available free-tier APIs. The resulting data then serve as the foundation for the subsequent integrity checks, point-in-time universe construction, data-quality procedures, and machine-learning pipeline.

11 U.S. Equity Market Proxy Selection

To construct a representative, investable, and unconfounded U.S. equity universe, this study selects the FTSE Russell 1000 Index as the primary market proxy. Comprising the 1,000 largest U.S.-domiciled securities ranked by total market capitalization within the Russell 3000E universe, the Russell 1000 captures approximately 90% to 95% of total investable U.S. equity capitalization while maintaining strict rules-based transparency [FTS26]. At any time t, constituents are weighted in proportion to their free-float-adjusted market capitalization $M_{i,t}^{\text{float}} = P_{i,t}(S_{i,t}^{\text{total}} - S_{i,t}^{\text{restricted}})$, yielding the index weight and effective diversification metric:

$$w_{i,t} = \frac{M_{i,t}^{\text{float}}}{\sum_{j=1}^{1000} M_{j,t}^{\text{float}}}, \qquad N_{\text{eff},t} = \frac{1}{\sum_{i=1}^{1000} w_{i,t}^2}. \tag{11.1}$$

While the top 200 mega-cap constituents account for roughly 78% of index capitalization (compressing $N_{\text{eff},t}$ and driving the primary market mode), the remaining 800 mid-cap constituents (representing $\approx 22\%$ of index weight, with a median capitalization of \$16.7 billion against a weighted average of \$1.56 trillion as of Q3 2026) provide substantial cross-sectional breadth [FTS26]. For cross-sectional ranking models, expanding the universe from 500 to 1,000 names increases the number of unique pairwise comparisons $\binom{N}{2} = \frac{N(N-1)}{2}$ fourfold, from 124,750 to 499,500 per period.

The empirical asset-pricing and market-microstructure literature provides three primary justifications for preferring the Russell 1000 over alternative benchmarks such as the S&P 500 or Russell 3000:

Elimination of Committee-Selection Bias

Unlike the S&P 500 – which relies on discretionary committee selection and enforces positive GAAP earnings screens – Russell 1000 membership is determined mechanically by market-capitalization rank on the prescribed rank day [CHL15;BDFM19]. As emphasized by [CHL15], discretionary S&P 500 additions correlate endogenously with firm profitability, visibility, and expected future performance, confounding cross-sectional return inferences. By contrast, the deterministic Russell 1000/2000 threshold at rank 1,000 creates exogenous variation in benchmark weights. Exploiting this cutoff in a fuzzy regression-discontinuity design (RDD), [CHL15] document that stocks migrating just above rank 1,000 into the top of the value-weighted Russell 2000 experience an approximate 7.3% jump in passive institutional demand and a 5.0% abnormal June return ($t = 2.65$) relative to near-identical stocks remaining at the bottom of the Russell 1000.

Well-Documented Reconstitution and Ownership Dynamics

Because Russell reconstitution follows a predictable schedule – historically annual in June, and transitioning to a semiannual June/December cycle in 2026 [FTS26] – its liquidity and ownership shocks are thoroughly parameterized in the literature. Early event studies by [Mad03], [BCJ04], [Che06], and [CH08] show that index rebalancing induces both temporary order-imbalance price pressure (e.g., a 10.78% June addition-minus-deletion spread followed by a −6.04% July reversal in [Mad03]) and permanent shifts in bid-ask spreads and institutional monitoring. Furthermore, [Pet11] quantifies the hidden rebalancing drag borne by passive funds during crowded index revisions, while [CP09] establish Russell indexes as foundational benchmarks for evaluating active management via Active Share ($\mathrm{AS}_t = \frac{1}{2}\sum_i |w_{i,t}^{\mathrm{fund}} - w_{i,t}|$). More recently, researchers have used Russell's mechanical assignment to identify how passive and ETF ownership improve corporate governance [AGK16], propagate return comovement [CSW16;DS18], and elevate daily volatility and short-horizon negative autocorrelation – yielding a risk premium of up to 56 basis points per month for high-ETF-ownership stocks [BDFM18].

Institutional Liquidity and Arbitrage Efficiency vs. the Russell 3000

While the broader Russell 3000 captures roughly 98% of the U.S. market by adding the 2,000 small-cap constituents of the Russell 2000 [FTS26], the bottom 2,000 names contribute only 3% to 8% of market capitalization while imposing severe implementation friction. Using Russell reconstitution as an instrument, [PA22] demonstrate that large- and mid-cap Russell 1000 securities exhibit low arbitrage constraints and unimpaired price discovery, whereas small- and micro-cap stocks suffer from binding short-sale and liquidity frictions. Under a standard square-root market-impact cost function for trade size $|Q_{i,t}|$ relative to average daily dollar volume $\mathrm{ADV}_{i,t}$ and volatility $\sigma_{i,t}$:

$$MI_{i,t} = \eta_i \sigma_{i,t} \left(\frac{|Q_{i,t}|}{\mathrm{ADV}_{i,t}} \right)^{1/2}, \tag{11.2}$$

truncating the universe at rank 1,000 avoids the steep market-impact decay, wide spreads, and elevated delisting bias of the small-cap tail while preserving the economically dominant core of the U.S. equity market.

12 Data Acquisition and Processing Pipeline

To construct a survivorship-bias-free dataset, we implement a three-stage data acquisition and normalization pipeline. Because standard financial databases frequently overwrite historical index memberships with current constituents, reconstructing point-in-time constituent lists is essential for empirical validity. Our workflow systematically captures annual snapshots of

the Russell 1000 Index following its late-June reconstitution and links these constituents to fundamental and market records.

12.1 Historical Index Constituent Extraction

The Russell 1000 Index undergoes an annual reconstitution on the final Friday of June, establishing the definitive constituent list for the subsequent twelve months. To capture true point-in-time index membership across our 2000–2026 sample period, we retrieve archival constituent snapshots directly from the Internet Archive's Wayback Machine immediately following each June reconstitution phase.

Because institutional data hosting and corporate ownership have migrated across platforms over the past two decades, we query multiple archival domains to ensure complete longitudinal coverage:

- **Official Index Providers:** Primary constituent lists and reconstitution reports published across current and legacy provider portals, including London Stock Exchange Group, FTSE Russell, Russell Investments and russell.com, and the defunct early-2000s Russell/Mellon analytical services domain.
- **Institutional ETF Holdings:** Post-reconstitution holdings disclosures from BlackRock for the iShares Russell 1000 ETF (IWB), which serve as a high-fidelity proxy and cross-validation source when direct provider tables are unavailable.

Due to changes in web standards over the 2000–2026 sample horizon, the 27 extracted raw files span heterogeneous file architectures, including unstructured Portable Document Format (`.pdf`) fact sheets, Microsoft Excel (`.xls`) and comma-separated values (`.csv`) spreadsheets, and JavaScript Object Notation (`.json`) API payloads (summarized in Table 12.1). We implement a custom parsing pipeline to strip document metadata, headers, and footnotes, standardizing each annual extraction into a clean, line-separated plain-text file containing company names and primary ticker symbols for each reconstitution year t.

Table 12.1: Raw Russell 1000 Constituent Snapshots

File Format	Count	Reconstitution Cohort Years	Primary Source Structure
PDF Document	21	2000–2007, 2009–2018, 2020–2022	Multi-page provider fact sheets
JSON Payload	3	2019, 2023, 2024	Structured web API responses
Excel Workbook	2	2025, 2026	Tabular holdings spreadsheets
CSV File	1	2008	Delimited constituent table
Total	**27**	**2000–2026 (Complete Panel)**	**Normalized to `.txt`**

12.2 WRDS Query Execution and Historical Data Retrieval

With the standardized annual constituent lists established, the second phase of the pipeline extracts longitudinal stock records via the Wharton Research Data Services web query interface, specifically targeting the Center for Research in Security Prices (CRSP) Stock Database (`crsp_a_stock`). To preserve the annual reconstitution structure and prevent look-ahead or survivorship bias, we execute separate cohort-specific queries for each reconstitution cycle rather than pooling tickers across the entire sample period.

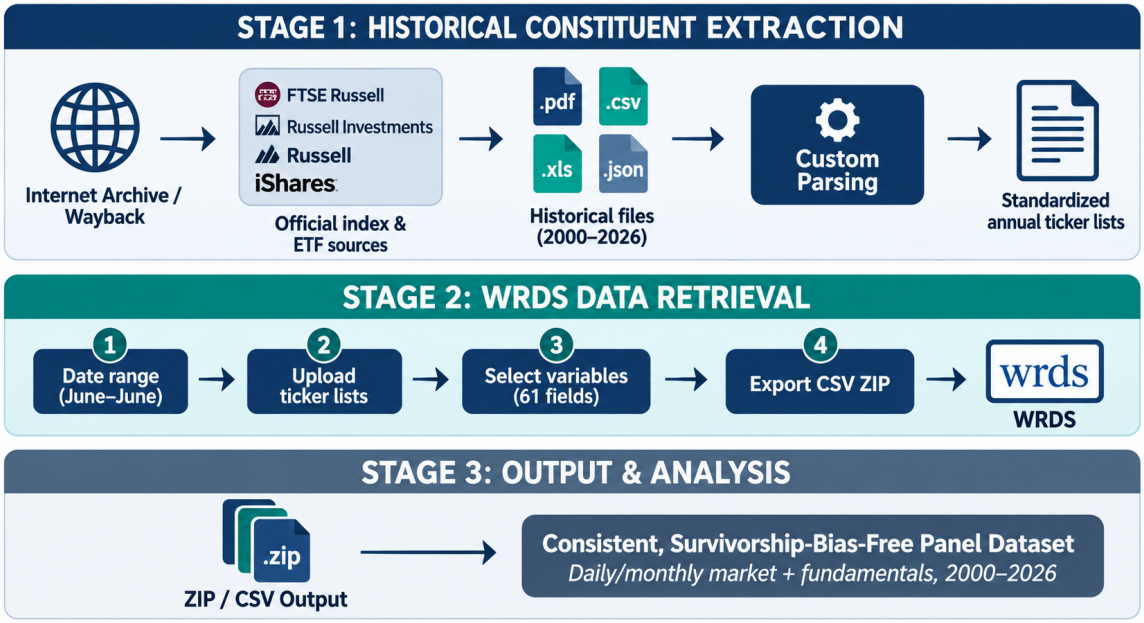


Figure 12.1: Historical Constituent Data Pipeline.

For each reconstitution year t, the WRDS web query execution follows a standardized four-step protocol:

Step 1 **Date Range:** Select the 12-month active cycle following reconstitution, from late June of year t to late June of year $t+1$ (e.g., `2023-06-30` to `2024-06-30`).

Step 2 **Code Upload:** Set the format to ticker symbols and upload the standardized plain-text file (`.txt`, one ticker per line) using the WRDS interface.

Step 3 **Variables:** Select 61 Identifying and Time Series variables to capture market data and security-level metadata.

Step 4 **Output:** Export the query results as comma-delimited text (`.csv`) in a compressed `.zip` archive using `YYYY-MM-DD` dates, then download directly from the WRDS portal.

13 Data Integrity

Part IV
Data Transformation

14 Data Sources and Preprocessing

14.1 Feature Engineering

14.1.1 Logarithmic Returns

Let $P_{i,t}$ denote the dividend-adjusted closing price of security i at time t. The single-period logarithmic return, $r_{i,t}$, is defined as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) = \ln(P_{i,t}) - \ln(P_{i,t-1}). \tag{14.1}$$

This transformation represents the continuously compounded return over a single period. Unlike simple returns, logarithmic returns are additive through time, such that the cumulative logarithmic return over multiple periods is equal to the sum of the corresponding single-period returns. This property provides a convenient representation for constructing rolling transformations from sequential price observations.

Because the transformation is based on relative rather than absolute price changes, logarithmic returns are independent of the nominal price level of a security. This provides a comparable representation of price movements across securities with substantially different prices and avoids directly modeling the non-stationary scale of raw price levels.

Within the data-processing pipeline, logarithmic returns therefore provide the baseline representation of short-horizon price dynamics. Subsequent transformations are constructed from this representation where appropriate.

14.1.2 Distance from Simple Moving Average (DSMA)

Let $SMA_{i,t}^{(w)}$ denote the Simple Moving Average of the closing price of security i over a rolling window of w days:

$$SMA_{i,t}^{(w)} = \frac{1}{w}\sum_{k=0}^{w-1} P_{i,t-k}, \tag{14.2}$$

where $P_{i,t}$ denotes the dividend-adjusted closing price. The Distance from Simple Moving Average (DSMA) measures the proportional deviation of the current price from its recent moving average:

$$DSMA_{i,t}^{(w)} = \frac{P_{i,t} - SMA_{i,t}^{(w)}}{SMA_{i,t}^{(w)}}. \tag{14.3}$$

In this study, the lookback windows are defined as $w \in \{10, 20, 60\}$ to represent short-, medium-, and long-horizon deviations from the corresponding local price baseline.

DSMA represents the relative position of the current price with respect to its recent moving average. Positive values indicate that the current price is above the corresponding moving average, while negative values indicate that it is below. Expressing the deviation proportionally rather than in absolute price units makes the resulting transformation comparable across securities with different nominal price levels.

As a data transformation, DSMA therefore converts the price level into a normalized measure of relative price position across multiple temporal horizons, providing information complementary to period-to-period returns.

14.1.3 Relative Strength Index (RSI)

Let $\Delta P_{i,t} = P_{i,t} - P_{i,t-1}$ denote the single-period change in the dividend-adjusted closing price of security i. The price changes are decomposed into positive gains, $U_{i,t}$, and absolute losses, $D_{i,t}$:

$$U_{i,t} = \max(\Delta P_{i,t}, 0), \quad D_{i,t} = \max(-\Delta P_{i,t}, 0). \tag{14.4}$$

For a lookback window of w periods, the Relative Strength factor is defined as the ratio of the smoothed average gain to the smoothed average loss:

$$RS_{i,t}^{(w)} = \frac{\text{AvgGain}_{i,t}^{(w)}}{\text{AvgLoss}_{i,t}^{(w)}}. \tag{14.5}$$

The Relative Strength Index is then given by

$$RSI_{i,t}^{(w)} = 100 - \frac{100}{1 + RS_{i,t}^{(w)}}. \tag{14.6}$$

The resulting indicator is bounded on $[0, 100]$. Higher values indicate that recent gains have been larger and/or more persistent relative to recent losses, whereas lower values indicate the opposite. The ratio-based construction therefore provides a common numerical representation of recent directional momentum across securities.

Within the data-processing pipeline, RSI transforms sequential price changes into a bounded representation of the relative balance between recent gains and losses. It thus provides information complementary to logarithmic returns and DSMA, which represent individual price changes and relative price position, respectively.

14.1.4 Relative Volume (RVOL)

Let $V_{i,t}$ denote the raw trading volume of security i at time t. Relative Volume, $RVOL_{i,t}^{(w)}$, measures current trading activity relative to the security's recent historical volume baseline.

The baseline is defined as

$$SMA_{V,i,t}^{(w)} = \frac{1}{w}\sum_{k=0}^{w-1} V_{i,t-k}, \tag{14.7}$$

and Relative Volume is given by

$$RVOL_{i,t}^{(w)} = \frac{V_{i,t}}{SMA_{V,i,t}^{(w)}}. \tag{14.8}$$

A value near 1 indicates that current trading volume is close to its recent average, whereas values substantially above or below 1 indicate unusually high or low activity relative to the security's own historical baseline. By expressing volume relative to its own recent level, the transformation reduces the dependence of the representation on absolute trading volume.

As a data transformation, RVOL converts raw volume into a relative measure of trading intensity. This provides a representation of changes in market activity that is more comparable across securities than absolute volume and complements price-derived transformations by incorporating information from the volume dimension of the OHLCV data.

14.1.5 Ichimoku Indicator

The Ichimoku Kinko Hyo (hereafter, Ichimoku) is a multi-component technical analysis framework that represents price structure across multiple temporal horizons. It combines rolling high–low price ranges into several related reference levels, allowing the relative position and configuration of price to be represented jointly [Cah12;DYW+21].

Let $P_{i,t}^H$, $P_{i,t}^L$, and $P_{i,t}^C$ denote the high, low, and closing prices of security i at time t. The framework uses three lookback parameters, w_1, w_2, and w_3.

Tenkan-sen The Tenkan-sen ($TS_{i,t}$) is the midpoint of the highest high and lowest low over the short-term window:

$$TS_{i,t} = \frac{\max_{k=0,\ldots,w_1-1} P_{i,t-k}^H + \min_{k=0,\ldots,w_1-1} P_{i,t-k}^L}{2}. \tag{14.9}$$

It provides a short-horizon reference level derived from the recent trading range and therefore represents local price structure over the corresponding temporal window.

Kijun-sen The Kijun-sen ($KS_{i,t}$) applies the same midpoint construction over the longer window w_2:

$$KS_{i,t} = \frac{\max_{k=0,\ldots,w_2-1} P_{i,t-k}^H + \min_{k=0,\ldots,w_2-1} P_{i,t-k}^L}{2}. \tag{14.10}$$

Because it is calculated over a longer horizon, the Kijun-sen provides a slower-moving reference for the prevailing price structure and complements the shorter-horizon Tenkan-sen.

Senkou Span A Senkou Span A (SSA) is defined as the midpoint of the Tenkan-sen and Kijun-sen:

$$SSA_{i,t+w_2} = \frac{TS_{i,t} + KS_{i,t}}{2}. \tag{14.11}$$

It combines short- and medium-horizon reference levels into a single structural representation.

Senkou Span B Senkou Span B (SSB) is the midpoint of the highest high and lowest low over the longest lookback window:

$$SSB_{i,t+w_2} = \frac{\max_{k=0,\ldots,w_3-1} P^H_{i,t-k} + \min_{k=0,\ldots,w_3-1} P^L_{i,t-k}}{2}. \tag{14.12}$$

It provides a slower-moving reference based on the longer historical trading range.

Kumo The Kumo, or cloud, is the region bounded by Senkou Span A and Senkou Span B. It therefore represents the relationship between the short/medium-horizon structure captured by SSA and the longer-horizon structure captured by SSB [Cah12;DYW+21].

Chikou Span The Chikou Span (CS) represents the current closing price plotted w_2 periods backward:

$$CS_{i,t-w_2} = P^C_{i,t}. \tag{14.13}$$

It provides a lagged comparison between current price and historical price levels.

Temporal Alignment Because Senkou spans and the Chikou Span are conventionally shifted for visualization, their use in a predictive setting requires point-in-time alignment. Features supplied to the model at time t must be constructed exclusively from information available no later than t. A forward-shifted value that depends on observations unavailable at time t would introduce look-ahead bias. The exact alignment procedure must therefore be defined according to the implemented data pipeline.

Collectively, the Ichimoku components provide a multi-horizon representation of price structure that complements return-, moving-average-, momentum-, and volume-based transformations.

14.1.6 Price Volume Trend (PVT)

Let $P_{i,t}$ and $V_{i,t}$ denote the dividend-adjusted closing price and trading volume, respectively, of security i at time t. The Price Volume Trend (PVT) is a cumulative price–volume transformation defined recursively as

$$PVT_{i,t} = PVT_{i,t-1} + V_{i,t}\left(\frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}\right). \tag{14.14}$$

The change in PVT is therefore proportional to both the relative price movement and the contemporaneous trading volume. Positive price changes increase the cumulative value, whereas

negative price changes decrease it. Unlike volume-only transformations, PVT incorporates the magnitude of the price movement into each volume contribution.

As a data transformation, PVT converts the separate price and volume observations into a cumulative price–volume representation. It therefore captures information about the historical interaction between price changes and trading activity that is not represented by logarithmic returns or Relative Volume alone.

Because PVT is cumulative and unbounded, its numerical scale depends on the historical path and volume scale of each security. Any subsequent normalization or transformation should therefore be documented separately from the mathematical definition of PVT.

14.1.7 Rolling Volatility

Let $r_{i,t}$ denote the single-period logarithmic return of security i at time t. Rolling Volatility, $\sigma_{i,t}^{(w)}$, measures the dispersion of returns over a historical lookback window of w periods. The rolling mean return is defined as

$$\bar{r}_{i,t}^{(w)} = \frac{1}{w} \sum_{k=0}^{w-1} r_{i,t-k}, \tag{14.15}$$

and rolling volatility is calculated as the sample standard deviation:

$$\sigma_{i,t}^{(w)} = \sqrt{\frac{1}{w-1} \sum_{k=0}^{w-1} \left(r_{i,t-k} - \bar{r}_{i,t}^{(w)} \right)^2}. \tag{14.16}$$

The resulting quantity represents the historical variability of returns around their local mean. Larger values indicate greater recent return dispersion, whereas smaller values indicate comparatively stable return behavior. Because the calculation is performed on logarithmic returns rather than absolute price levels, the resulting measure is independent of the nominal price scale of the security.

As a data transformation, rolling volatility summarizes the variability of individual return observations over a defined historical horizon. It therefore complements logarithmic returns by representing the local dispersion of price movements rather than the movement of price itself.

For reproducibility, the implementation must specify the lookback window, whether the volatility measure is retained as a daily standard deviation or annualized, and the degrees-of-freedom convention used in the calculation.

14.2 Feature Extraction

To be updated



14.3 Feature Selection

To be updated

Part V

Alpha Implementation

15 Model Architecture

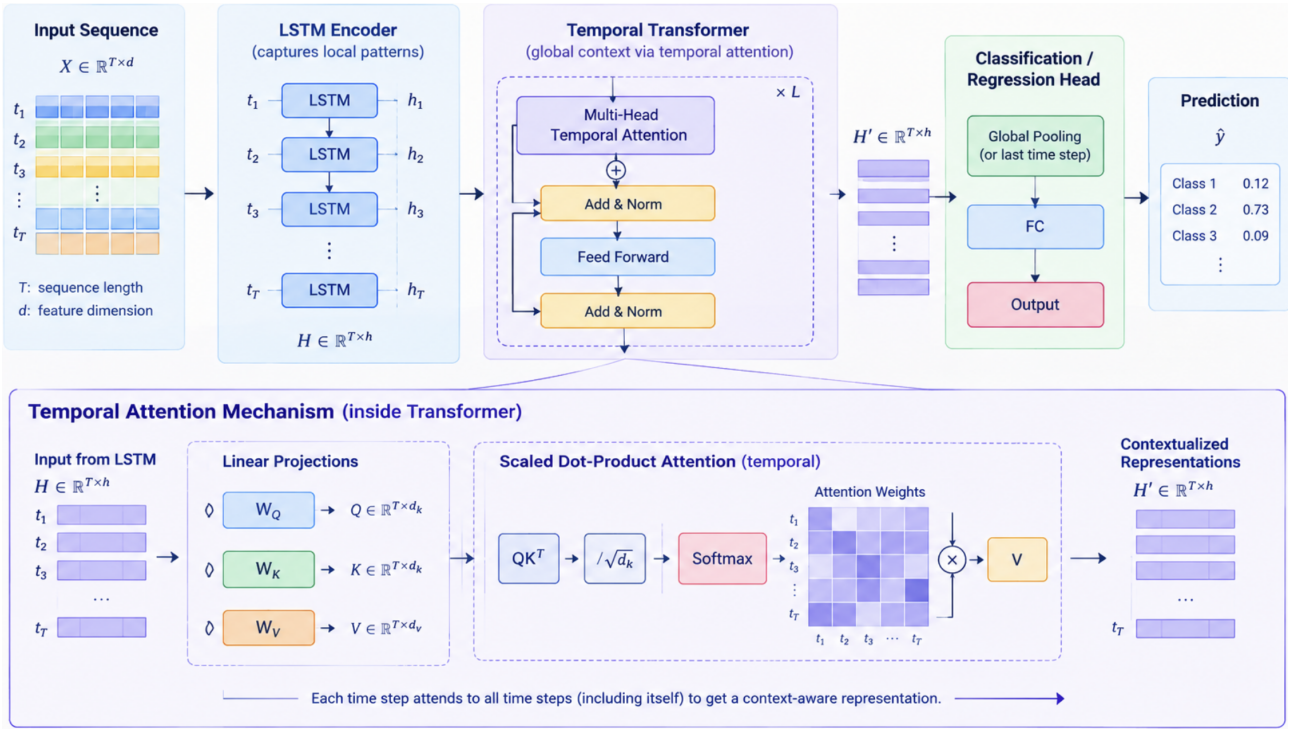


Figure 15.1: LSTM-Transformer Architecture

15.1 Temporal Encoding (LSTM)

15.2 Context Layer (Cross-Sectional Transformer)

15.3 Hybrid Loss Function

16 Hyperparameter Optimization

17 Training Procedure



Part VI
Empirical Results

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Part VII

Limitations and Discussion

Part VIII

Conclusion

Appendices

Appendix A Nested Walk-Forward Cross-Validation

Deep learning architectures, such as Long Short-Term Memory (LSTM) networks, Temporal Convolutional Networks (TCNs), and Spatio-Temporal Graph Neural Networks (ST-GNNs), are increasingly applied to complex panel datasets. These datasets, common in quantitative finance, econometrics, epidemiology, and meteorology, consist of multiple entities (the cross-section) observed over multiple time steps.

When training these high-capacity models, practitioners face two primary forms of data leakage:

1. **Temporal leakage (look-ahead bias):** utilizing future information to predict past or present states.
2. **Cross-sectional leakage:** normalizing or scaling data using aggregate statistics that include testing periods across different entities.

To rigorously evaluate generalization, the validation scheme must respect the strict arrow of time while accommodating the cross-sectional dimension. The Nested Walk-Forward Cross-Sectional Validation method is designed for this purpose, providing a conservative and realistic estimate of how a model will perform in a sequential production environment. The approach combines rolling-origin evaluation for time-series forecasting[HA21] with the principle that hyperparameter selection and final performance estimation must be performed on separate data[CT10].

A.1 Theoretical Foundation

Standard Validation versus Temporal Realities In standard machine learning, k-fold cross-validation randomly partitions the dataset into k subsets, iteratively using $k-1$ folds for training and one for testing. For panel data, random shuffling destroys temporal autocorrelation and allows the model to access information from future periods. Even grouped k-fold validation, which keeps observations from the same entity together, does not respect the temporal ordering of the data and therefore cannot account for regime shifts or non-stationarity over time. In financial

applications, these violations are especially consequential because overlapping forward-looking labels and low signal-to-noise ratios can produce misleadingly strong in-sample results [LdP18].

Table A.1: Comparison of Validation Methods for Panel Data

Method	Time-respected?	Unbiased HP tuning?	Cross-section intact?	Cost
Standard k-fold	No	No	Broken	Low
Grouped k-fold	No	No	Intact	Low
Simple walk-forward	Yes	No	Intact	Medium
Nested walk-forward	**Yes**	**Yes**	**Intact**	**High**

Walk-Forward Validation Walk-forward validation, also known as rolling-origin evaluation or backtesting, ensures that the training set strictly precedes the validation or test set chronologically [Par08]. Let the dataset be indexed by time $t \in \{1, 2, \ldots, T\}$. A walk-forward setup evaluates the model at time t using only data from 1 to $t-1$ (expanding window) or from $t-w$ to $t-1$ (rolling window of size w). This structure emulates the repeated retraining and prediction process that would occur in live deployment.

The Necessity of the Nested Approach Deep learning models possess numerous hyperparameters, including learning rate, dropout rate, number of layers, embedding dimensions, and window lengths. If a simple walk-forward procedure uses the validation period both to select hyperparameters and to report final performance, the resulting metrics are optimistically biased. This phenomenon, known as *overfitting to the validation set*, arises because repeated model selection on the same validation data effectively incorporates information about that data into the modeling pipeline [CT10].

A nested approach addresses this problem by strictly separating hyperparameter selection from out-of-sample performance estimation through two distinct, temporally ordered loops:

- **Inner loop (hyperparameter optimization):** This loop operates entirely within the training window of the current outer step. It performs its own chronological walk-forward validation across multiple historical folds. For a given hyperparameter configuration, the model is trained and evaluated sequentially, and the average validation performance determines the configuration's fitness. Because every inner validation set precedes the corresponding outer test set, the selected hyperparameters contain no look-ahead information about the subsequent out-of-sample period [CT10;KK22].
- **Outer loop (unbiased evaluation):** The outer loop simulates sequential deployment over time. Once the inner loop identifies the best hyperparameter configuration, a new model is trained using that configuration on all historical data available up to the outer

test date. The model is then evaluated on strictly future observations. Advancing the outer window through time produces a sequence of genuinely out-of-sample predictions, whose aggregation provides a realistic estimate of end-to-end pipeline performance[Par08;HA21].

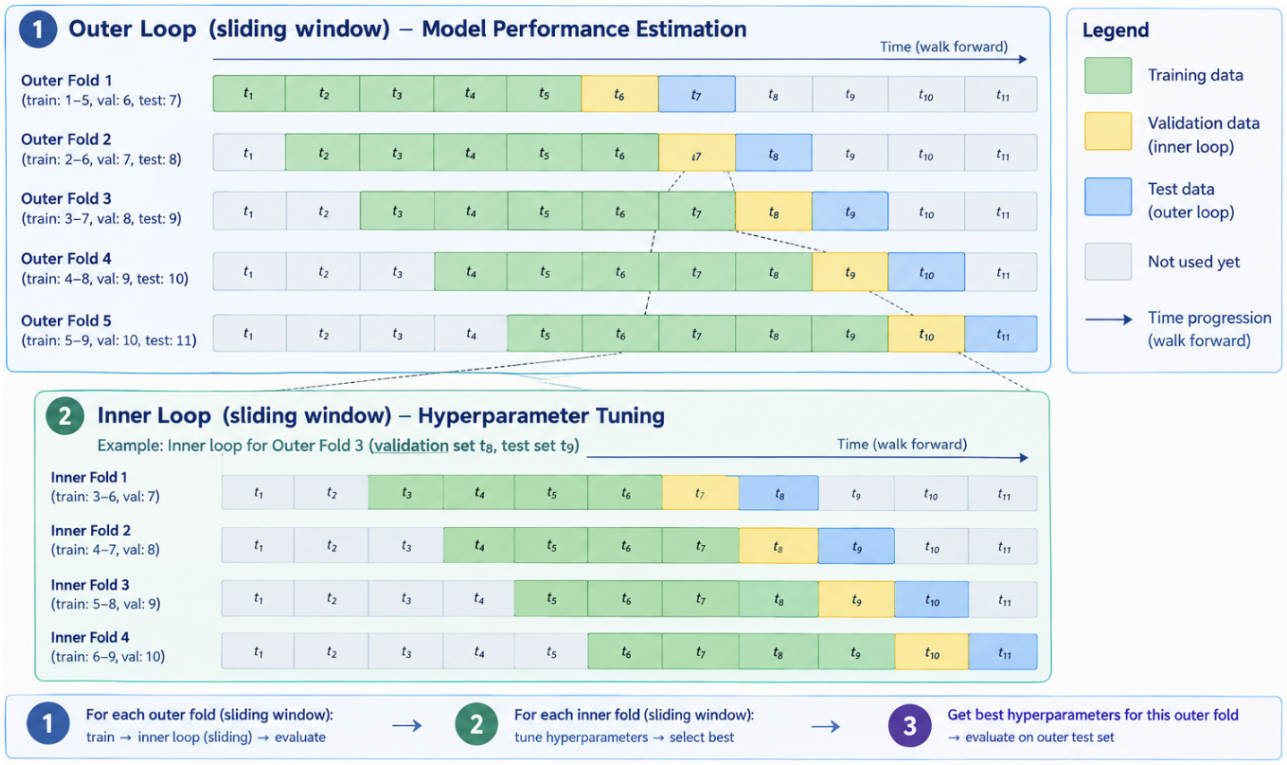


Figure A.1: Walk-Forward Cross-Sectional Validation Visualization

Leakage Control for Overlapping Financial Labels

When the target variable is a multi-period forward return or another overlapping label, temporal separation of training and test dates alone may not eliminate leakage. In that case, training observations whose label horizons overlap the validation or test interval should be removed through purging, and an embargo period may be added after the training window to attenuate serial dependence across adjacent periods[LdP18]. These safeguards should be applied consistently in both the inner and outer loops. A nested walk-forward implementation of this type has been used in cross-sectional equity-return prediction, where an outer walk-forward loop evaluates performance year by year while an inner leave-one-year-out procedure selects the regularization parameter using only prior-year data[KK22].

A.2 Formal Specification

A.2.1 Mathematical Formulation

Algorithm 1 Nested Walk-Forward Cross-Sectional Validation Pseudocode

Require: Dataset $\mathcal{D}$ sorted by time t, hyperparameter search space H, initial training window W_{init}, step size S.

$E_{\text{OOS}} \leftarrow []$ ▷ Out-of-sample errors
for $T_{\text{test}} = W_{\text{init}}$ **to** T **step** S **do**
  $\mathcal{D}_{\text{outer_train}} \leftarrow \mathcal{D}[: T_{\text{test}} - 1]$
  $\mathcal{D}_{\text{outer_test}} \leftarrow \mathcal{D}[T_{\text{test}} : T_{\text{test}} + S - 1]$
  $best_h \leftarrow$ null, $min_val_loss \leftarrow \infty$
  for $h \in H$ **do** ▷ Inner loop: hyperparameter tuning
    $val_loss \leftarrow 0$
    Define chronological inner validation splits within $\mathcal{D}_{\text{outer_train}}$
    for each inner split $(\mathcal{D}_{\text{inner_train}}, \mathcal{D}_{\text{inner_val}})$ **do**
      $\theta \leftarrow$ InitializeWeights()
      $\theta \leftarrow$ TrainDeepModel$(\mathcal{D}_{\text{inner_train}}, \theta, h)$
      $val_loss \leftarrow val_loss +$ EvaluateCrossSection$(\mathcal{D}_{\text{inner_val}}, \theta)$
    end for
    $val_loss \leftarrow val_loss/$number_of_inner_splits
    if $val_loss < min_val_loss$ **then**
      $min_val_loss \leftarrow val_loss$
      $best_h \leftarrow h$
    end if
  end for
  ▷ Outer loop: final training and evaluation
  $\theta_{\text{final}} \leftarrow$ InitializeWeights()
  $\theta_{\text{final}} \leftarrow$ TrainDeepModel$(\mathcal{D}_{\text{outer_train}}, \theta_{\text{final}}, best_h)$
  $err \leftarrow$ EvaluateCrossSection$(\mathcal{D}_{\text{outer_test}}, \theta_{\text{final}})$
  E_{OOS}.append(err)
end for
return Mean(E_{OOS})

Consider a panel dataset $\mathcal{D} = \{(\mathbf{x}_{i,t}, y_{i,t})\}$, where $i \in \{1, \ldots, N\}$ denotes the cross-sectional entity and $t \in \{1, \ldots, T\}$ denotes the time step. Here, $\mathbf{x}_{i,t} \in \mathbb{R}^d$ is the feature vector and $y_{i,t}$ is the target variable.

Let H be the set of candidate hyperparameter configurations, and let the outer test dates be $T_{\text{test}}^{(1)}, T_{\text{test}}^{(2)}, \ldots, T_{\text{test}}^{(K)}$.

For the k-th outer fold at time $T_{\text{test}}^{(k)}$:

1. **Outer training window:** The available data are
$$\mathcal{D}_{\text{train}}^{(k)} = \left\{ (\mathbf{x}_{i,t}, y_{i,t}) \mid t < T_{\text{test}}^{(k)} \right\}. \tag{A.1}$$
2. **Inner loop (hyperparameter optimization):** Split $\mathcal{D}_{\text{train}}^{(k)}$ into further chronological sub-windows. Let the inner test dates be $\tau^{(1)}, \tau^{(2)}, \ldots, \tau^{(J)}$, where $\tau^{(j)} < T_{\text{test}}^{(k)}$. For each

hyperparameter configuration $h \in H$,

$$\text{Score}(h) = \frac{1}{J}\sum_{j=1}^{J} \mathcal{L}\left(\mathcal{M}_h\left(\mathcal{D}_{\text{train},\tau^{(j)}}\right), \mathcal{D}_{\text{test},\tau^{(j)}}\right), \tag{A.2}$$

where $\mathcal{M}_h$ is the model trained with hyperparameters h and $\mathcal{L}$ is a cross-sectional loss, such as mean squared error or cross-entropy computed across all N entities at time $\tau^{(j)}$. The selected configuration is

$$h_{\text{opt}}^{(k)} = \arg\min_{h\in H} \text{Score}(h). \tag{A.3}$$

3. **Outer loop (evaluation):** The final model is trained on the full outer training window using the selected configuration:

$$\mathcal{M}_{\text{final}}^{(k)} = \text{Train}\left(\mathcal{D}_{\text{train}}^{(k)}, h_{\text{opt}}^{(k)}\right). \tag{A.4}$$

Its out-of-sample performance is evaluated on the cross-section at $T_{\text{test}}^{(k)}$:

$$\text{Error}^{(k)} = \mathcal{L}\left(\mathcal{M}_{\text{final}}^{(k)}, \left\{\left(\mathbf{x}_{i,T_{\text{test}}^{(k)}}, y_{i,T_{\text{test}}^{(k)}}\right)\right\}_{i=1}^{N}\right). \tag{A.5}$$

The final estimate of generalization performance is obtained by aggregating $\text{Error}^{(k)}$ across all outer folds. Because each evaluation uses a model trained and tuned exclusively on earlier data, the resulting metric reflects the expected performance of the complete modeling pipeline under sequential deployment [HA21;Par08].

A.2.2 Quantitative Finance Mathematical Formulation

To adapt Nested Walk-Forward Cross-subSectional Validation to systematic trading and quantitative portfolio construction, market microstructure dynamics and risk constraints are explicitly formalized below.

Cross-Sectional Alpha Optimization Let $\mathbf{r}_{t+1} \in \mathbb{R}^N$ denote the forward return vector for an asset universe $\mathcal{U} = \{1, \ldots, N\}$ at timestamp t. The neural network $f_\theta : \mathbb{R}^{N\times d} \to \mathbb{R}^N$ maps asset features $\mathbf{X}_t$ to predicted alpha scores $\hat{\mathbf{r}}_{t+1} = f_\theta(\mathbf{X}_t)$.

Definition 1 (Cross-Sectional Z-Score Operator)**.** For any vector $\mathbf{v} \in \mathbb{R}^N$, the cross-sectional normalization operator $\mathbf{Z} : \mathbb{R}^N \to \mathbb{R}^N$ is defined as:

$$\mathbf{Z}(\mathbf{v}) = \frac{v - \bar{v}}{\sigma_{\mathbf{v}}}, \quad \bar{v} = \frac{1}{N}\sum_{i=1}^{N} v_i, \quad \sigma_{\mathbf{v}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(v_i - \bar{v})^2} \tag{A.6}$$

Definition 2 (Differentiable Information Coefficient Loss)**.** The network optimizes the cross-sectional Pearson correlation (Information Coefficient) between predicted rankings and realized forward returns:

$$\mathcal{L}_{\text{IC}}(\theta; \mathbf{X}_t, \mathbf{r}_{t+1}) = -\frac{1}{N}\mathbf{Z}\left(f_\theta(\mathbf{X}_t)\right)^\top \mathbf{Z}\left(\mathbf{r}_{t+1}\right) \tag{A.7}$$

Purging and Embargo Operators For an h-period forward return target $y_{i,t} = \prod_{k=1}^{h}(1 + r_{i,t+k}) - 1$, overlapping target horizons introduce non-zero temporal covariance across model residuals.

Definition 3 (Information Leakage Sets)**.** Given an inner validation set $\mathcal{T}_{\text{val}} = [t_{\text{v,start}}, t_{\text{v,end}}]$ and an embargo horizon $e \geq h$:

$$\text{Purged Set: } \mathcal{P}(\mathcal{T}_{\text{val}}) = \{t \mid t_{\text{v,start}} - h + 1 \leq t < t_{\text{v,start}}\} \tag{A.8}$$

$$\text{Embargo Set: } \mathcal{E}(\mathcal{T}_{\text{val}}, e) = \{t \mid t_{\text{v,end}} < t \leq t_{\text{v,end}} + e\} \tag{A.9}$$

The strictly admissible training set $\mathcal{T}_{\text{train}}^{\text{admissible}}$ is given by:

$$\mathcal{T}_{\text{train}}^{\text{admissible}} = \mathcal{T}_{\text{hist}} \setminus (\mathcal{T}_{\text{val}} \cup \mathcal{P}(\mathcal{T}_{\text{val}}) \cup \mathcal{E}(\mathcal{T}_{\text{val}}, e)) \tag{A.10}$$

Inner Loop Hyperparameter Selection Objective To avoid validation overfit under a low signal-to-noise ratio (SNR $\ll 1$), the inner-loop hyperparameter fitness function $g(h)$ evaluates risk-adjusted portfolio performance across J inner folds:

$$h_{\text{opt}}^{(k)} = \arg\max_{h \in H} \frac{1}{J} \sum_{j=1}^{J} \left[\frac{\mu_{\text{IC}}(h,j)}{\sigma_{\text{IC}}(h,j) + \epsilon} + \lambda \cdot \text{Sharpe}\left(\mathbf{w}_t^*(f_{\theta_{h,j}})\right) \right] \tag{A.11}$$

where portfolio weights $\mathbf{w}_t^* \in \mathbb{R}^N$ solve the mean-variance transaction-cost constrained problem:

$$\mathbf{w}_t^*(\hat{\mathbf{r}}_{t+1}) = \arg\max_{\mathbf{w} \in \mathcal{W}} \left(\mathbf{w}^\top \hat{\mathbf{r}}_{t+1} - \frac{\gamma}{2} \mathbf{w}^\top \boldsymbol{\Sigma}_t \mathbf{w} - \kappa \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \right) \tag{A.12}$$

Regime Drift Regularization via Proximal Updates To reduce computational burden while adapting to market regime changes, outer model weights $\theta^{(k)}$ undergo proximal updating from step $k-1$:

$$\theta^{(k)} = \arg\min_{\theta} \left(\sum_{t \in \mathcal{D}_{\text{outer}}^{(k)}} \mathcal{L}_{\text{IC}}(\theta; \mathbf{X}_t, \mathbf{r}_{t+1}) + \frac{\mu}{2} \left\| \theta - \theta^{(k-1)} \right\|_{\mathbf{F}}^2 \right) \tag{A.13}$$

where $\|\mathbf{v}\|_{\mathbf{F}}^2 = \mathbf{v}^\top \mathbf{F} \mathbf{v}$, and $\mathbf{F} = \mathbb{E}\left[\nabla_\theta \mathcal{L} \left(\nabla_\theta \mathcal{L} \right)^\top \right]$ is the diagonal Fisher Information Matrix.

A.3 Conclusion

The Nested Walk-Forward Cross-Sectional Validation method is a rigorous framework for developing deep learning models on time-series panel data, where temporal leakage can invalidate performance estimates. By structurally separating hyperparameter search in the inner loop from temporal out-of-sample evaluation in the outer loop, and by evaluating errors across the cross-sectional dimension, it provides robust and realistic performance metrics [CT10;Par08]. Although computationally expensive, the procedure is particularly valuable in applications such as algorithmic trading, macroeconomic forecasting, epidemiology, and personalized medicine, where deployment decisions depend on reliable estimates of future performance [LdP18;HA21].

Appendix B Performance Framework and Alpha Metrics

B.1 Information Efficiency Metrics

Information Coefficient (IC)

Information Coefficient Information Ratio (ICIR)

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